

LAB 1

מספר זוג:

12

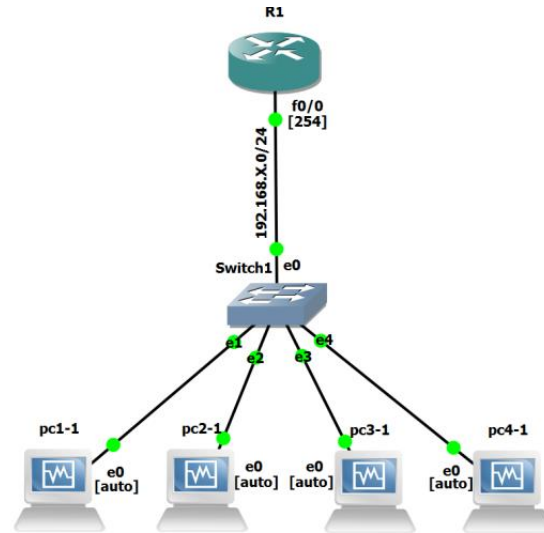
מגישים:

שיר משה 318492667

עידו בן בעש"ט 316415843

1.DHCP (Dynamic Host Configuration Protocol)

Topology:



1.1 – 1.5 Configure Router:

R1#configure terminal
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.12.254 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end

1.6 Configuring the DHCP Server:

R1(config)#ip dhcp pool 1
R1(dhcp-config)#network 192.168.12.0 /24
R1(dhcp-config)#domain-name Lan1
R1(dhcp-config)#default-router 192.168.12.254
R1(dhcp-config)#lease 0 0 2

1.8

show ip dhcp pool show ip dhcp bind
show ip dhcp bind

```
R1#show ip dhcp pool
Pool 1 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 0
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index    IP address range    Leased addresses
    192.168.12.1     192.168.12.1 - 192.168.12.254    0
R1#show ip dhcp bind
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
R1#
```

Configuring the DHCP Clients:

1.9 Setup up a DHCP client on each of the PCs, as follows:

```
nano /etc/network/interfaces
```

נרשום את השורות הבאות בקובץ :

auto eth0
iface eth0 inet dhcp

(ctr X and save)

1.10. Reboot all the PCs

1.11 Write down the IP address and default gateway of each PC:

PC1

```
root@pc1:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:50:3d:86
          inet addr:192.168.12.3  Bcast:192.168.12.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:744 (744.0 B)  TX bytes:684 (684.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:16 errors:0 dropped:0 overruns:0 frame:0
          TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1184 (1.1 KB)  TX bytes:1184 (1.1 KB)

root@pc1:~# route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
default    192.168.12.254 0.0.0.0 UG 0 0 0 eth0
192.168.12.0 * 255.255.255.0 U 0 0 0 eth0
root@pc1:~#
```

PC2

```
root@pc2:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:f8:60:55
          inet addr:192.168.12.4  Bcast:192.168.12.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:6 errors:0 dropped:0 overruns:0 frame:0
          TX packets:4 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1206 (1.2 KB)  TX bytes:1086 (1.0 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:16 errors:0 dropped:0 overruns:0 frame:0
          TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1184 (1.1 KB)  TX bytes:1184 (1.1 KB)

root@pc2:~# route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
default    192.168.12.254 0.0.0.0 UG 0 0 0 eth0
192.168.12.0 * 255.255.255.0 U 0 0 0 eth0
root@pc2:~# _
```

PC3

```
root@pc3:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d1:e4:56
          inet addr:192.168.12.1  Bcast:192.168.12.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:20 errors:0 dropped:0 overruns:0 frame:0
          TX packets:15 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4302 (4.3 KB)  TX bytes:3438 (3.4 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:16 errors:0 dropped:0 overruns:0 frame:0
          TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1184 (1.1 KB)  TX bytes:1184 (1.1 KB)

root@pc3:~# route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
default    192.168.12.254 0.0.0.0 UG 0 0 0 eth0
192.168.12.0 * 255.255.255.0 U 0 0 0 eth0
root@pc3:~# _
```

PC4:

```
root@pc4:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:b8:e6:de
          inet addr:192.168.12.2  Bcast:192.168.12.255  Mask:255.255.255.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:22 errors:0 dropped:0 overruns:0 frame:0
          TX packets:20 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4422 (4.4 KB)  TX bytes:4302 (4.3 KB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:16 errors:0 dropped:0 overruns:0 frame:0
          TX packets:16 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1184 (1.1 KB)  TX bytes:1184 (1.1 KB)

root@pc4:~# route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
default    192.168.12.254 0.0.0.0 UG 0 0 0 eth0
192.168.12.0 * 255.255.255.0 U 0 0 0 eth0
root@pc4:~#
```

1.12

After each of the PCs gets an IP address, run again the following commands on R1 and save the output:

```
show ip dhcp pool
```

```
show ip dhcp bind
```

```
R1#show ip dhcp pool

Pool 1 :
  Utilization mark (high/low)      : 100 / 0
  Subnet size (first/next)         : 0 / 0
  Total addresses                   : 254
  Leased addresses                  : 4
  Pending event                    : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.12.5   192.168.12.1 - 192.168.12.254  4
R1#show ip dhcp bind
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
192.168.12.1    0800.27d1.e456   Mar 01 2002 12:45 AM Automatic
192.168.12.2    0800.27b8.e6de   Mar 01 2002 12:45 AM Automatic
192.168.12.3    0800.2750.3d86   Mar 01 2002 12:45 AM Automatic
192.168.12.4    0800.27f8.6055   Mar 01 2002 12:45 AM Automatic
R1#
```

Report

1.15

R1(config)#ip dhcp pool 1	מגדירים עבודה עם DHCP ומספר POOL
R1(dhcp-config)#network 192.168.12.0 /24	מגדירים מרחב כתובות IP עם subnetmask
R1(dhcp-config)#domain-name Lan1	להגדיר לרשת את השם שלה
R1(dhcp-config)#default-router 192.168.12.254	להגדיר את הראוטר הראשי ב-LAN
R1(dhcp-config)#lease 0 0 2	אחרי כל 2 דקות הלקוחות מבקשים אוטומטית את הכתובת שוב ושוב

1.16

clientn הוא הראשון שפנה לבקש כתובת מהראוטר הראשי, הוא זה שמתחיל את סשן ההודעות של ה-DHCP:

128	14:19:47.687536	0.0.0.0	255.255.255.255	DHCP	342	0	✓	DHCP Discover
130	14:19:49.712413	192.168.12.254	192.168.12.1	DHCP	342	1	✓	DHCP Offer
131	14:19:49.713098	0.0.0.0	255.255.255.255	DHCP	342	0	✓	DHCP Request
132	14:19:49.806994	192.168.12.254	192.168.12.1	DHCP	342	1	✓	DHCP ACK

1.17 הראוטר בוחר את הכתובות שהוא מחלק לפי הסדר. הכתובות רשומות בתמונות ב 1.11

1.18

1. הודעת Discover – המחשב שרוצה לקבל כתובת IP מהראוטר הראשי ב LAN, שולח הודעה עם כתובת מקור 0.0.0.0 בברודקאסט כדי לחפש מי הראוטר הראשי.

2. הודעת Offer – הראוטר שולח הודעה עם כתובת IP שהוא מציע למחשב.

3. הודעת Request – המחשב מחליט אם הוא מקבל את הכתובת שהוצעה לו ושולח הודעת סבבה בברודקאסט.

4. הודעת ACK – הראוטר מאשר סופית את הכתובת ללקוח.

1.19 בפעם הראשונה לא היו שום מחשבים שמחוברים (שרואים ע"י BIND), בפעם השניה רואים את הרשימה של כל 4 המחשבים שמחוברים.

רואים בפעם הראשונה leased address = 0 כי לא היו שום כתובות שחולקו, ובפעם השניה 4.

רואים שבפעם השניה הכתובת הבאה שמחלקים היא עם אוקטטה 5 בסוף. כי זה לפי הסדר מתחלק.

```
R1#show ip dhcp pool
Pool 1 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 0
  Pending event : none
  1 subnet is currently in the pool :
    Current index IP address range Leased addresses
    192.168.12.1 192.168.12.1 - 192.168.12.254 0
R1#show ip dhcp bind
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ User name Lease expiration Type
R1#
```

```
R1#show ip dhcp pool
Pool 1 :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 254
  Leased addresses : 4
  Pending event : none
  1 subnet is currently in the pool :
    Current index IP address range Leased addresses
    192.168.12.5 192.168.12.1 - 192.168.12.254 4
R1#show ip dhcp bind
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ User name Lease expiration Type
192.168.12.1 0800.27d1.e456 Mar 01 2002 12:45 AM Automatic
192.168.12.2 0800.27b8.e6de Mar 01 2002 12:45 AM Automatic
192.168.12.3 0800.2750.3d86 Mar 01 2002 12:45 AM Automatic
192.168.12.4 0800.27f8.6055 Mar 01 2002 12:45 AM Automatic
R1#
```

1.20

יש הודעות שכל דקה נשלחות. הודעות Request , Ack. (חצי מהזמן שהוגדר לזריקה מהרשת) כל דקה

484	14:38:31.470961	192.168.12.1	192.168.12.254	DHCP	342	1 ✓	DHCP Request
485	14:38:31.480265	192.168.12.254	192.168.12.1	DHCP	342	1 ✓	DHCP ACK
486	14:38:32.992169	192.168.12.3	192.168.12.254	DHCP	342	3 ✓	DHCP Request
487	14:38:33.000144	192.168.12.254	192.168.12.3	DHCP	342	3 ✓	DHCP ACK
493	14:38:39.958376	192.168.12.2	192.168.12.254	DHCP	342	2 ✓	DHCP Request
494	14:38:39.969179	192.168.12.254	192.168.12.2	DHCP	342	2 ✓	DHCP ACK
495	14:38:40.459917	192.168.12.4	192.168.12.254	DHCP	342	4 ✓	DHCP Request
496	14:38:40.464256	192.168.12.254	192.168.12.4	DHCP	342	4 ✓	DHCP ACK
506	14:39:23.047030	192.168.12.1	192.168.12.254	DHCP	342	1 ✓	DHCP Request

2. DHCP Static Mapping

2.2 bootp בwireshark

2.3 נעשה פקודה של ifconfig במחשב 1 כדי למצוא את הMAC

2.4 תצוגת הMAC

2.5 כתובת IP - 192.168.12.3

MAC – 08:00:27:50:3d:86

2.6

R1#configure terminal
R1(config)#ip dhcp pool STATIC
R1(dhcp-config)#host 192.168.12.42 /24
R1(dhcp-config)#hardware-address 0800.2750.3d86
R1(dhcp-config)#end

2.7 לחכות 2 דקות

2.8

show ip dhcp pool
show ip dhcp bind

```

R1#show ip dhcp pool

Pool 1 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses               : 4
  Pending event                  : none
  1 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.12.5   192.168.12.1 - 192.168.12.254  4

Pool STATIC :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 1
  Leased addresses               : 1
  Pending event                  : none
  0 subnet is currently in the pool :
    Current index  IP address range  Leased addresses
    192.168.12.42  192.168.12.42 - 192.168.12.42  1

R1#show ip dhcp bind
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration   Type
Hardware address/
User name
192.168.12.1    0800.27d1.e456   Mar 01 2002 01:27 AM Automatic
192.168.12.2    0800.27b8.e6de   Mar 01 2002 01:27 AM Automatic
192.168.12.3    0800.2750.3d86   Mar 01 2002 01:27 AM Automatic
192.168.12.4    0800.27f9.6055   Mar 01 2002 01:27 AM Automatic
192.168.12.42   0800.2750.3d86   Infinite           Manual

```

2.10

```

R1#show ip dhcp pool

Pool 1 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 3
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.12.5 192.168.12.1 - 192.168.12.254 3

Pool STATIC :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 1
Leased addresses : 1
Pending event : none
0 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.12.42 192.168.12.42 - 192.168.12.42 1

R1#show ip dhcp bind
Bindings from all pools not associated with VRF:
IP address Client-ID/ Hardware address/ User name Lease expiration Type
192.168.12.1 0800.27d1.e456 Mar 01 2002 01:31 AM Automatic
192.168.12.2 0800.27b8.e6de Mar 01 2002 01:31 AM Automatic
192.168.12.4 0800.27f8.6055 Mar 01 2002 01:32 AM Automatic
192.168.12.42 0800.2750.3d86 Infinite Manual

```

2.11

```

root@pc1:~# ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:50:3d:86
          inet addr:192.168.12.42 Bcast:192.168.12.255 Mask:255.255.255.0

```

Report

2.13

R1#configure terminal	כניסה לשלב של הגדרות הראוטר
R1(config)#ip dhcp pool STATIC	הגדרה של כתובת בצורה ידנית
R1(dhcp-config)#host 192.168.12.42 /24	כתובת ה IP שנרצה להגדיר
R1(dhcp-config)#hardware-address 0800.2750.3d86	כתובת ה MAC של המחשב שלו נרצה להגדיר IP חדש ידנית
R1(dhcp-config)#end	סיום הקונפיגורציה

2.14 לפני הריסט היינו מחוברים גם אוטומטית וגם ידנית, אפשר לראות 2 כתובות IP בטבלה.

אח"כ רק ידנית. נשים לב שהוא אינפניטי- כלומר לא צריך את הזמן שחרור שיש לאוטומטיים.

2.15 לפני זה היו 5 IP 4 מחשבים, אחכ 4 ו 4 .

2.16 כשינינו את ההגדרות בראוטר, רק התווספה שורה בטבלה עם כתובת IP ונוספת לאותו מחשב.
רק לאחר שעשינו reboot והמחשב שחרר את הכתובת IP האוטומטית נשארו עם הידנית.

2.17 רואים ב wireshark, שיש הודעת release כאשר עושים reboot למחשב. הודעה זו אומרת שהמחשב כבה ולא צריך יותר כתובת IP. הפרוטוקול לא מחייב את הודעת השחרור, אם לא הייתה מגיעה כזאת, לאחר 2 דקות שלא שלחו הודעת request מאותו מחשב, הראוטר לבדו היה משחרר את הכתובת הזו מהמחשב.

4. VLAN (Virtual Local Area Network)

4.3. נקונפג את ה IP הסטטיים של המחשבים

```
ifconfig eth0 192.168.1.1 netmask 255.255.255.0 broadcast 192.168.1.255
```

4.5

ESW1#configure terminal
ESW1(config)#interface FastEthernet1/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan (X+1)
ESW1(config-if)#no shutdown
ESW1(config-if)#end

4.7.

נשלח פינג בין כל המחשבים: ping -c4 IP_ADDRESS,

PC1:

```
root@pc1:~# ping -c4 192.168.12.2
PING 192.168.12.2 (192.168.12.2) 56(84) bytes of data.
^[[AFrom 192.168.12.1 icmp_seq=1 Destination Host Unreachable
From 192.168.12.1 icmp_seq=2 Destination Host Unreachable
From 192.168.12.1 icmp_seq=3 Destination Host Unreachable
From 192.168.12.1 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3015ms
pipe 3
root@pc1:~# ping -c4 192.168.12.3
PING 192.168.12.3 (192.168.12.3) 56(84) bytes of data.
64 bytes from 192.168.12.3: icmp_seq=1 ttl=64 time=2.70 ms
64 bytes from 192.168.12.3: icmp_seq=2 ttl=64 time=1.70 ms
64 bytes from 192.168.12.3: icmp_seq=3 ttl=64 time=0.851 ms
64 bytes from 192.168.12.3: icmp_seq=4 ttl=64 time=0.889 ms

--- 192.168.12.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 0.851/1.536/2.701/0.753 ms
root@pc1:~# ping -c4 192.168.12.4
PING 192.168.12.4 (192.168.12.4) 56(84) bytes of data.
From 192.168.12.1 icmp_seq=1 Destination Host Unreachable
From 192.168.12.1 icmp_seq=2 Destination Host Unreachable
From 192.168.12.1 icmp_seq=3 Destination Host Unreachable
From 192.168.12.1 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.4 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3000ms
pipe 3
root@pc1:~# _
```

PC2:

```
root@pc2:~# ping -c4 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data.
From 192.168.12.2 icmp_seq=1 Destination Host Unreachable
From 192.168.12.2 icmp_seq=2 Destination Host Unreachable
From 192.168.12.2 icmp_seq=3 Destination Host Unreachable
From 192.168.12.2 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3016ms
pipe 3
root@pc2:~# ping -c4 192.168.12.3
PING 192.168.12.3 (192.168.12.3) 56(84) bytes of data.
From 192.168.12.2 icmp_seq=1 Destination Host Unreachable
From 192.168.12.2 icmp_seq=2 Destination Host Unreachable
From 192.168.12.2 icmp_seq=3 Destination Host Unreachable
From 192.168.12.2 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.3 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3015ms
pipe 3
root@pc2:~# ping -c4 192.168.12.4
PING 192.168.12.4 (192.168.12.4) 56(84) bytes of data.
64 bytes from 192.168.12.4: icmp_seq=1 ttl=64 time=2.44 ms
64 bytes from 192.168.12.4: icmp_seq=2 ttl=64 time=1.82 ms
64 bytes from 192.168.12.4: icmp_seq=3 ttl=64 time=1.67 ms
64 bytes from 192.168.12.4: icmp_seq=4 ttl=64 time=1.65 ms

--- 192.168.12.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.655/1.898/2.441/0.319 ms
root@pc2:~#
```

PC3:

```
root@pc3:~# ping -c4 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data:
64 bytes from 192.168.12.1: icmp_seq=1 ttl=64 time=1.12 ms
64 bytes from 192.168.12.1: icmp_seq=2 ttl=64 time=1.53 ms
64 bytes from 192.168.12.1: icmp_seq=3 ttl=64 time=1.76 ms
64 bytes from 192.168.12.1: icmp_seq=4 ttl=64 time=2.24 ms

--- 192.168.12.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.122/1.667/2.247/0.407 ms
root@pc3:~# ping -c4 192.168.12.2
PING 192.168.12.2 (192.168.12.2) 56(84) bytes of data:
From 192.168.12.3 icmp_seq=1 Destination Host Unreachable
From 192.168.12.3 icmp_seq=2 Destination Host Unreachable
From 192.168.12.3 icmp_seq=3 Destination Host Unreachable
From 192.168.12.3 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.2 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3000ms
pipe 3
root@pc3:~# ping -c4 192.168.12.4
PING 192.168.12.4 (192.168.12.4) 56(84) bytes of data:
From 192.168.12.3 icmp_seq=1 Destination Host Unreachable
From 192.168.12.3 icmp_seq=2 Destination Host Unreachable
From 192.168.12.3 icmp_seq=3 Destination Host Unreachable
From 192.168.12.3 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.4 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 2999ms
pipe 3
root@pc3:~#
```

PC4:

```
root@pc4:~# ping -c4 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data:
From 192.168.12.4 icmp_seq=1 Destination Host Unreachable
From 192.168.12.4 icmp_seq=2 Destination Host Unreachable
From 192.168.12.4 icmp_seq=3 Destination Host Unreachable
From 192.168.12.4 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3015ms
pipe 3
root@pc4:~# ping -c4 192.168.12.2
PING 192.168.12.2 (192.168.12.2) 56(84) bytes of data:
64 bytes from 192.168.12.2: icmp_seq=1 ttl=64 time=1.62 ms
64 bytes from 192.168.12.2: icmp_seq=2 ttl=64 time=0.576 ms
64 bytes from 192.168.12.2: icmp_seq=3 ttl=64 time=1.78 ms
64 bytes from 192.168.12.2: icmp_seq=4 ttl=64 time=1.62 ms

--- 192.168.12.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 0.576/1.402/1.780/0.481 ms
root@pc4:~# ping -c4 192.168.12.3
PING 192.168.12.3 (192.168.12.3) 56(84) bytes of data:
From 192.168.12.4 icmp_seq=1 Destination Host Unreachable
From 192.168.12.4 icmp_seq=2 Destination Host Unreachable
From 192.168.12.4 icmp_seq=3 Destination Host Unreachable
From 192.168.12.4 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.3 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3002ms
pipe 3
root@pc4:~#
```

4.8

שלוח 4 פיגים מ־PC1 לכתובת IP:

ping -c4 192.168.12.5

```
root@pc1:~# ping -c4 192.168.12.5
connect: Network is unreachable
```

Report

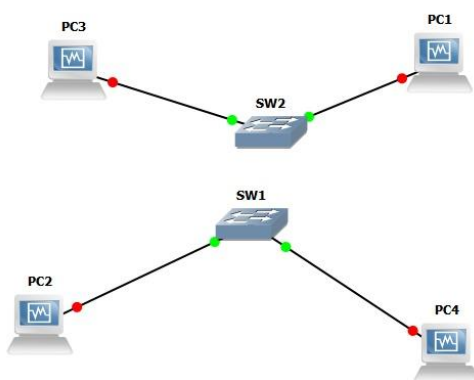
4.10

ראינו שהצלחנו לשלוח פינגים כמו שצריך רק בין מחשבים תחת אותו VLAN

	PC1	PC2	PC3	PC4
PC1	-	4	0	0
PC2	4	-	0	0
PC3	0	0	-	4
PC4	0	0	4	-

4.11 המחשב ששולח משתמש באינטרפייס שלו שמחובר לראוטר הראשי, הראוטר מחפש בטבלת ה VLAN של אותו PC (13VLAN) יש שם רק את אינטרפייס של מחשב 3. וזהו.

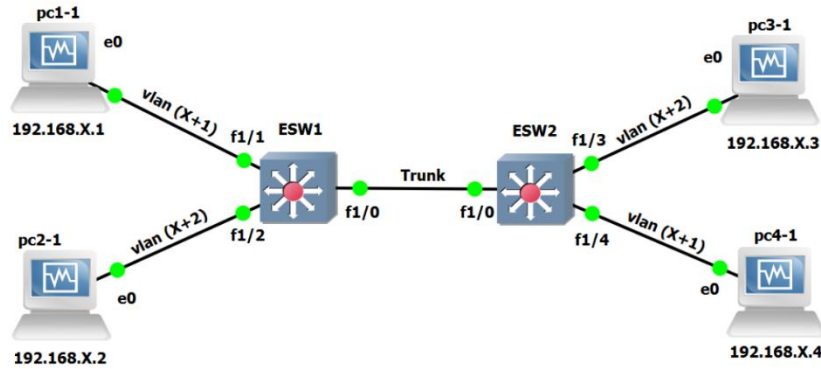
4.12 נייצר VLAN 2 ע"י שימוש בswitch רגיל בצורה הבאה:



4.13 הודעות STP הן של spanning tree protocol שמודיעות על מחשב חדש שהצטרף לרשת, ונשלחות באופן מחזורי כדי שידעו איזה מחשב עדיין קיים ברשת (keep alive).

5. VLAN DOT1Q (IEEE 802.1Q) Trunk

5.1 נבנה את הטופולוגיה הבאה: (כולל קינפוג סטטי של כתובות IP של המחשבים)



5.2 נקנפג את הסוויצ'ים:

ESW1#configure terminal
ESW1(config)#interface FastEthernet1/0
ESW1(config-if)# switchport mode trunk
ESW1(config-if)# switchport trunk encapsulation dot1Q
ESW1(config-if)#no shutdown
ESW1(config-if)#end

5.3

נקנפג את הפורטים שמחוברים למחשבים לפי הפורטים הרלוונטים:

ESW1#configure terminal
ESW1(config)#interface FastEthernet1/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan (X+1)
ESW1(config-if)#no shutdown
ESW1(config-if)#end

5.4 נפתח wireshark בכל אחד מהפורטים

5.5 נשלח 4 פיגים בין כל המחשבים IP_ADDRESS -c4 ping

ולהלן התוצאות:

```
root@pc1:~# ping -c4 192.168.12.2
PING 192.168.12.2 (192.168.12.2) 56(84) bytes of data.
From 192.168.12.1 icmp_seq=1 Destination Host Unreachable
From 192.168.12.1 icmp_seq=2 Destination Host Unreachable
From 192.168.12.1 icmp_seq=3 Destination Host Unreachable
From 192.168.12.1 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.2 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 2999ms
pipe 3
root@pc1:~# ping -c4 192.168.12.3
PING 192.168.12.3 (192.168.12.3) 56(84) bytes of data.
From 192.168.12.1 icmp_seq=1 Destination Host Unreachable
From 192.168.12.1 icmp_seq=2 Destination Host Unreachable
From 192.168.12.1 icmp_seq=3 Destination Host Unreachable
From 192.168.12.1 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.3 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3008ms
pipe 3
root@pc1:~# ping -c4 192.168.12.4
PING 192.168.12.4 (192.168.12.4) 56(84) bytes of data.
64 bytes from 192.168.12.4: icmp_seq=1 ttl=64 time=3.86 ms
64 bytes from 192.168.12.4: icmp_seq=2 ttl=64 time=2.02 ms
64 bytes from 192.168.12.4: icmp_seq=3 ttl=64 time=1.84 ms
64 bytes from 192.168.12.4: icmp_seq=4 ttl=64 time=1.38 ms

--- 192.168.12.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.383/2.280/3.863/0.944 ms
root@pc1:~#
```

```
root@pc3:~# ping -c4 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data.
From 192.168.12.3 icmp_seq=1 Destination Host Unreachable
From 192.168.12.3 icmp_seq=2 Destination Host Unreachable
From 192.168.12.3 icmp_seq=3 Destination Host Unreachable
From 192.168.12.3 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3015ms
pipe 3
root@pc3:~# ping -c4 192.168.12.2
PING 192.168.12.2 (192.168.12.2) 56(84) bytes of data.
64 bytes from 192.168.12.2: icmp_seq=1 ttl=64 time=1.87 ms
64 bytes from 192.168.12.2: icmp_seq=2 ttl=64 time=2.22 ms
64 bytes from 192.168.12.2: icmp_seq=3 ttl=64 time=1.74 ms
64 bytes from 192.168.12.2: icmp_seq=4 ttl=64 time=1.68 ms

--- 192.168.12.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.680/1.882/2.229/0.218 ms
root@pc3:~# ping -c4 192.168.12.4
PING 192.168.12.4 (192.168.12.4) 56(84) bytes of data.
From 192.168.12.3 icmp_seq=1 Destination Host Unreachable
From 192.168.12.3 icmp_seq=2 Destination Host Unreachable
From 192.168.12.3 icmp_seq=3 Destination Host Unreachable
From 192.168.12.3 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.4 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3016ms
pipe 3
root@pc3:~#
```

```
root@pc2:~# ping -c4 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data.
From 192.168.12.2 icmp_seq=1 Destination Host Unreachable
From 192.168.12.2 icmp_seq=2 Destination Host Unreachable
From 192.168.12.2 icmp_seq=3 Destination Host Unreachable
From 192.168.12.2 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3017ms
pipe 3
root@pc2:~# ping -c4 192.168.12.3
PING 192.168.12.3 (192.168.12.3) 56(84) bytes of data.
64 bytes from 192.168.12.3: icmp_seq=1 ttl=64 time=3.17 ms
64 bytes from 192.168.12.3: icmp_seq=2 ttl=64 time=1.56 ms
64 bytes from 192.168.12.3: icmp_seq=3 ttl=64 time=1.83 ms
64 bytes from 192.168.12.3: icmp_seq=4 ttl=64 time=1.28 ms

--- 192.168.12.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.289/1.966/3.176/0.726 ms
root@pc2:~# ping -c4 192.168.12.4
PING 192.168.12.4 (192.168.12.4) 56(84) bytes of data.
From 192.168.12.2 icmp_seq=1 Destination Host Unreachable
From 192.168.12.2 icmp_seq=2 Destination Host Unreachable
From 192.168.12.2 icmp_seq=3 Destination Host Unreachable
From 192.168.12.2 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.4 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3002ms
pipe 3
root@pc2:~#
```

```
root@pc4:~# ping -c4 192.168.12.1
PING 192.168.12.1 (192.168.12.1) 56(84) bytes of data.
64 bytes from 192.168.12.1: icmp_seq=1 ttl=64 time=3.46 ms
64 bytes from 192.168.12.1: icmp_seq=2 ttl=64 time=1.88 ms
64 bytes from 192.168.12.1: icmp_seq=3 ttl=64 time=1.90 ms
64 bytes from 192.168.12.1: icmp_seq=4 ttl=64 time=2.12 ms

--- 192.168.12.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.881/2.343/3.465/0.655 ms
root@pc4:~# ping -c4 192.168.12.2
PING 192.168.12.2 (192.168.12.2) 56(84) bytes of data.
From 192.168.12.4 icmp_seq=1 Destination Host Unreachable
From 192.168.12.4 icmp_seq=2 Destination Host Unreachable
From 192.168.12.4 icmp_seq=3 Destination Host Unreachable
From 192.168.12.4 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.2 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3015ms
pipe 3
root@pc4:~# ping -c4 192.168.12.3
PING 192.168.12.3 (192.168.12.3) 56(84) bytes of data.
From 192.168.12.4 icmp_seq=1 Destination Host Unreachable
From 192.168.12.4 icmp_seq=2 Destination Host Unreachable
From 192.168.12.4 icmp_seq=3 Destination Host Unreachable
From 192.168.12.4 icmp_seq=4 Destination Host Unreachable

--- 192.168.12.3 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3002ms
pipe 3
root@pc4:~#
```

5.6 נעצור את ה wireshark

5.7 נריץ את הפקודה הבאה על ESW1: show vlan-switch

```
ESW1#show vlan-switch
```

VLAN Name	Status	Ports
1 default	active	Fal/3, Fal/4, Fal/5, Fal/6 Fal/7, Fal/8, Fal/9, Fal/10 Fal/11, Fal/12, Fal/13, Fal/14 Fal/15
13 VLAN0013	active	Fal/1
14 VLAN0014	active	Fal/2
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	1002	1003
13	enet	100013	1500	-	-	-	-	-	0	0
14	enet	100014	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	1	1003
1003	tr	101003	1500	1005	0	-	-	srbr	1	1002
1004	fdnet	101004	1500	-	-	1	-	ibm	0	0
1005	trnet	101005	1500	-	-	1	-	ibm	0	0

```
ESW1#
```

REPORT

5.8

ESW1#configure terminal	כניסה לטרמינל
ESW1(config)#interface FastEthernet1/0	להדליק פורט 1/0
ESW1(config-if)# switchport mode trunk	קינפוג הפורט למצב שיכול לנהל תעבורה מכמה VLAN
ESW1(config-if)# switchport trunk encapsulation dot1Q	מאפשר לפורט להכניס TAG לפריים (שידע לאיזה VLAN לשייך)
ESW1(config-if)#no shutdown	
ESW1(config-if)#end	

5.9

	PC1	PC2	PC3	PC4
PC1	-	0	0	4
PC2	0	-	4	0
PC3	0	4	-	0
PC4	4	0	0	-

5.10

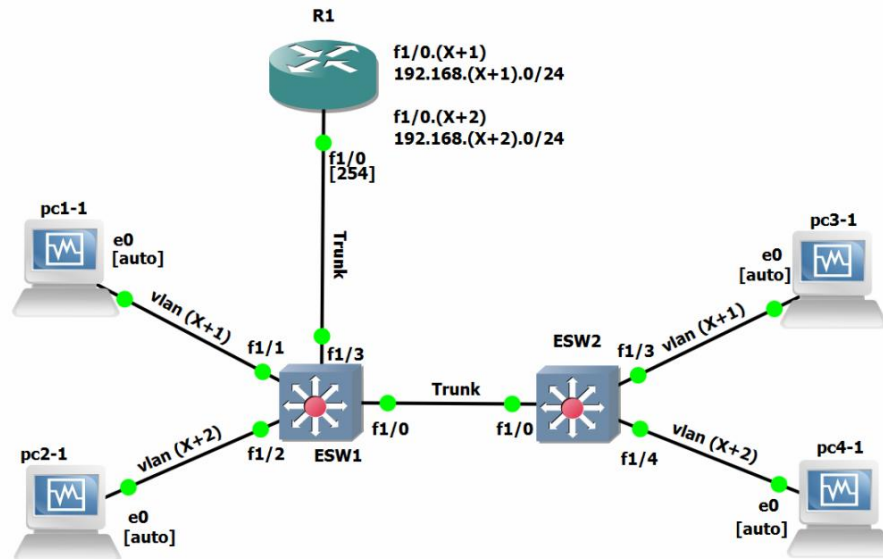
```
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 13
  000. .... = Priority: Best Effort (default) (0)
  ...0 .... = DEI: Ineligible
  .... 0000 0000 1101 = ID: 13
  Type: IPv4 (0x0800)
```

show vlan-switch 5.11

ע"י הרצה של הפקודה הזו אפשר לראות את הVLAN שמחוברים לאותו switch, את המספר שלהם, את השם שלהם, את הסטטוס (אם אקטיביים)

6.Router - VLAN and DHCP:

6.1 נבנה את הטופולוגיה הבאה:



6.2 נגדיר למחשבים לעבוד ב DHCP

nano /etc/network/interfaces
auto eth0
iface eth0 inet dhcp

6.3 נגדיר את הפורטים של הסוויצ'ים

נקנפג את הפורטים של ה trunk : (כולל את 1/3 שמחובר לראוטר)

ESW1#configure terminal
ESW1(config)#interface FastEthernet1/0
ESW1(config-if)# switchport mode trunk
ESW1(config-if)# switchport trunk encapsulation dot1Q
ESW1(config-if)#no shutdown
ESW1(config-if)#end

נקנפג את הפורטים שמחוברים למחשבים:

ESW1#configure terminal
ESW1(config)#interface FastEthernet1/1
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan (X+1)
ESW1(config-if)#no shutdown
ESW1(config-if)#end

6.4 נקנפג את הראוטר עבור VALN 13

R1#configure terminal
R1(config)# interface FastEthernet1/0
R1(config-if)#no shutdown
R1(config-if)#end
R1#configure terminal
R1(config)#interface FastEthernet1/0.13
R1(config-subif)#encapsulation dot1Q 13
R1(config-subif)#ip address 192.168.13.254 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#end

6.5 נקנפג את הראוטר עבור VALN 14

R1#configure terminal
R1(config)# interface FastEthernet1/0
R1(config-if)#no shutdown
R1(config-if)#end
R1#configure terminal
R1(config)#interface FastEthernet1/0.14
R1(config-subif)#encapsulation dot1Q 14
R1(config-subif)#ip address 192.168.14.254 255.255.255.0
R1(config-subif)#no shutdown

6.6 נקפג את הראוטר לעבוד ב DHCP עבור 2 ה LAN

R1#configure terminal
R1(config)#ip dhcp pool 1
R1(dhcp-config)#network 192.168.12.0 /24
R1(dhcp-config)#domain-name Lan1
R1(dhcp-config)#default-router 192.168.12.254
R1(dhcp-config)#lease 0 0 2

R1#configure terminal
R1(config)#ip dhcp pool 2
R1(dhcp-config)#network 192.168.13.0 /24
R1(dhcp-config)#domain-name Lan2
R1(dhcp-config)#default-router 192.168.13.254
R1(dhcp-config)#lease 0 0 2

6.7 נפתח wireshark על כל הפורטים

6.8 ping -c4 IP_ADDRESS מכולם פניגים מכולם לכולם

```
root@pc1:~# ping -c4 192.168.13.2
PING 192.168.13.2 (192.168.13.2) 56(84) bytes of data.
64 bytes from 192.168.13.2: icmp_seq=1 ttl=64 time=1.10 ms
64 bytes from 192.168.13.2: icmp_seq=2 ttl=64 time=1.34 ms
64 bytes from 192.168.13.2: icmp_seq=3 ttl=64 time=1.70 ms
64 bytes from 192.168.13.2: icmp_seq=4 ttl=64 time=1.34 ms

--- 192.168.13.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.100/1.372/1.700/0.213 ms
root@pc1:~# ping -c4 192.168.14.2
PING 192.168.14.2 (192.168.14.2) 56(84) bytes of data.
64 bytes from 192.168.14.2: icmp_seq=1 ttl=63 time=21.2 ms
64 bytes from 192.168.14.2: icmp_seq=2 ttl=63 time=20.9 ms
64 bytes from 192.168.14.2: icmp_seq=3 ttl=63 time=15.6 ms
64 bytes from 192.168.14.2: icmp_seq=4 ttl=63 time=13.2 ms

--- 192.168.14.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 13.279/17.761/21.233/3.422 ms
root@pc1:~# ping -c4 192.168.14.1
PING 192.168.14.1 (192.168.14.1) 56(84) bytes of data.
64 bytes from 192.168.14.1: icmp_seq=1 ttl=63 time=15.3 ms
64 bytes from 192.168.14.1: icmp_seq=2 ttl=63 time=12.1 ms
64 bytes from 192.168.14.1: icmp_seq=3 ttl=63 time=21.8 ms
64 bytes from 192.168.14.1: icmp_seq=4 ttl=63 time=15.9 ms

--- 192.168.14.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 12.101/16.302/21.831/3.512 ms
root@pc1:~# _
```

```
root@pc3:~# ping -c4 192.168.14.1
PING 192.168.14.1 (192.168.14.1) 56(84) bytes of data.
64 bytes from 192.168.14.1: icmp_seq=1 ttl=63 time=18.3 ms
64 bytes from 192.168.14.1: icmp_seq=2 ttl=63 time=18.4 ms
64 bytes from 192.168.14.1: icmp_seq=3 ttl=63 time=21.5 ms
64 bytes from 192.168.14.1: icmp_seq=4 ttl=63 time=19.6 ms

--- 192.168.14.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 18.336/19.497/21.583/1.325 ms
root@pc3:~# ping -c4 192.168.14.2
PING 192.168.14.2 (192.168.14.2) 56(84) bytes of data.
64 bytes from 192.168.14.2: icmp_seq=1 ttl=63 time=21.2 ms
64 bytes from 192.168.14.2: icmp_seq=2 ttl=63 time=14.0 ms
64 bytes from 192.168.14.2: icmp_seq=3 ttl=63 time=15.1 ms
64 bytes from 192.168.14.2: icmp_seq=4 ttl=63 time=14.4 ms

--- 192.168.14.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 14.057/16.233/21.236/2.917 ms
root@pc3:~# ping -c4 192.168.13.1
PING 192.168.13.1 (192.168.13.1) 56(84) bytes of data.
64 bytes from 192.168.13.1: icmp_seq=1 ttl=64 time=1.66 ms
64 bytes from 192.168.13.1: icmp_seq=2 ttl=64 time=1.32 ms
64 bytes from 192.168.13.1: icmp_seq=3 ttl=64 time=1.34 ms
64 bytes from 192.168.13.1: icmp_seq=4 ttl=64 time=1.41 ms

--- 192.168.13.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.324/1.437/1.667/0.137 ms
root@pc3:~#
```

```
root@pc2:~# ping -c4 192.168.13.1
PING 192.168.13.1 (192.168.13.1) 56(84) bytes of data.
64 bytes from 192.168.13.1: icmp_seq=1 ttl=63 time=17.5 ms
64 bytes from 192.168.13.1: icmp_seq=2 ttl=63 time=19.6 ms
64 bytes from 192.168.13.1: icmp_seq=3 ttl=63 time=21.6 ms
64 bytes from 192.168.13.1: icmp_seq=4 ttl=63 time=18.7 ms

--- 192.168.13.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 17.516/19.385/21.606/1.494 ms
root@pc2:~# ping -c4 192.168.13.2
PING 192.168.13.2 (192.168.13.2) 56(84) bytes of data.
64 bytes from 192.168.13.2: icmp_seq=1 ttl=63 time=14.9 ms
64 bytes from 192.168.13.2: icmp_seq=2 ttl=63 time=17.0 ms
64 bytes from 192.168.13.2: icmp_seq=3 ttl=63 time=21.3 ms
64 bytes from 192.168.13.2: icmp_seq=4 ttl=63 time=20.5 ms

--- 192.168.13.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 13.369/16.491/20.544/2.687 ms
root@pc2:~# ping -c4 192.168.14.2
PING 192.168.14.2 (192.168.14.2) 56(84) bytes of data.
64 bytes from 192.168.14.2: icmp_seq=1 ttl=64 time=1.59 ms
64 bytes from 192.168.14.2: icmp_seq=2 ttl=64 time=1.77 ms
64 bytes from 192.168.14.2: icmp_seq=3 ttl=64 time=1.16 ms
64 bytes from 192.168.14.2: icmp_seq=4 ttl=64 time=1.43 ms

--- 192.168.14.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.163/1.489/1.776/0.229 ms
root@pc2:~# _
```

```
root@pc4:~# ping -c4 192.168.14.1
PING 192.168.14.1 (192.168.14.1) 56(84) bytes of data.
64 bytes from 192.168.14.1: icmp_seq=1 ttl=64 time=3.00 ms
64 bytes from 192.168.14.1: icmp_seq=2 ttl=64 time=2.00 ms
64 bytes from 192.168.14.1: icmp_seq=3 ttl=64 time=1.64 ms
64 bytes from 192.168.14.1: icmp_seq=4 ttl=64 time=1.84 ms

--- 192.168.14.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.643/2.123/3.001/0.523 ms
root@pc4:~# ping -c4 192.168.13.1
PING 192.168.13.1 (192.168.13.1) 56(84) bytes of data.
64 bytes from 192.168.13.1: icmp_seq=1 ttl=63 time=22.5 ms
64 bytes from 192.168.13.1: icmp_seq=2 ttl=63 time=15.6 ms
64 bytes from 192.168.13.1: icmp_seq=3 ttl=63 time=14.3 ms
64 bytes from 192.168.13.1: icmp_seq=4 ttl=63 time=21.1 ms

--- 192.168.13.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 14.310/18.394/22.517/3.494 ms
root@pc4:~# ping -c4 192.168.13.2
PING 192.168.13.2 (192.168.13.2) 56(84) bytes of data.
64 bytes from 192.168.13.2: icmp_seq=1 ttl=63 time=21.9 ms
64 bytes from 192.168.13.2: icmp_seq=2 ttl=63 time=15.5 ms
64 bytes from 192.168.13.2: icmp_seq=3 ttl=63 time=18.7 ms
64 bytes from 192.168.13.2: icmp_seq=4 ttl=63 time=22.2 ms

--- 192.168.13.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 15.541/19.635/22.234/2.731 ms
root@pc4:~# _
```

Report

6.10

R1#configure terminal	נתחיל לקנפג
R1(config)# interface FastEthernet1/0	נדליק את אינטרפייס 1/0
R1(config-if)#no shutdown	
R1(config-if)#end	
R1#configure terminal	
R1(config)#interface FastEthernet1/0.13	נגדיר את הVLAN באותו פורט
R1(config-subif)#encapsulation dot1Q 13	נגדיר את הפרוטוקול של חלוקת הכתובות
R1(config-subif)#ip address 192.168.13.254 255.255.255.0	תחום הכתובות של הDHCP של VLAN זה
R1(config-subif)#no shutdown	
R1(config-subif)#end	

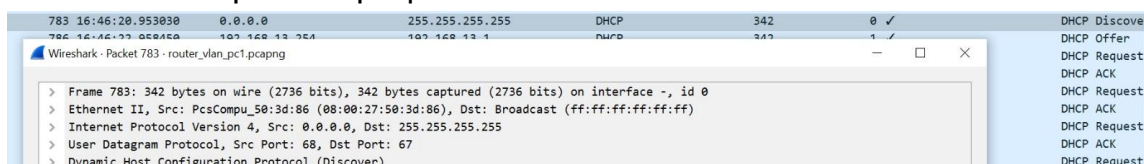
6.11

	PC1	PC2	PC3	PC4
PC1	-	4	4	4
PC2	4	-	4	4
PC3	4	4	-	4
PC4	4	4	4	-

הפעם כל הפינגים הצליחו להישלח בזכות הראוטר שיכול להעביר את התעבורה בין הרשתות כי הוא מכיר את כולן.

6.12 הראוטר הוא זה שמביא את כתובות ה IP ב DHCP, ולכן כל מחשב שמתחבר ומבקשה כתובת שולח הודעת discover בברודקסט שמגיעה גם לסוויצ' הקרוב שמעביר את ההודעה לראוטר ומשם חזרה דרך הסוויצ' למחשב. הברודקסט עובר על כל הלינקים ברשת מכיוון שאין PC1 עדיין רשת.

1 PC שולח broadcast של הודעת discover והיא עוברת בקו בין PC1 לבין 1S



ההודעה מ-1PC עוברת מהסוויץ' לראוטר:

472	16:46:21.562845	0.0.0.0	255.255.255.255	DHCP	346	0 ✓	DHCP Discover
477	16:46:22.958450	192.168.13.254	192.168.13.1	DHCP	346	1 ✓	DHCP Offer
479	16:46:22.959356	0.0.0.0	255.255.255.255	DHCP	346	0 ✓	DHCP Request

Wireshark - Packet 472 - router_vlan_r1.pcapng							
> Frame 472: 346 bytes on wire (2768 bits), 346 bytes captured (2768 bits) on interface -, id 0 > Ethernet II, Src: PcsCompu_f8:60:55 (08:00:27:f8:60:55), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 14 > Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255 > User Datagram Protocol, Src Port: 68, Dst Port: 67 > Dynamic Host Configuration Protocol (Discover)							

ההודעה עוברת בלינק trunk: (אמורה להגיע גם ל-3PC כי הוא גם תחת אותו broadcast).

2546	16:46:21.561868	0.0.0.0	255.255.255.255	DHCP	346	0 ✓	DHCP Discover
------	-----------------	---------	-----------------	------	-----	-----	---------------

Wireshark - Packet 2546 - router_vlan_ews1and2.pcapng							
> Frame 2546: 346 bytes on wire (2768 bits), 346 bytes captured (2768 bits) on interface -, id 0 > Ethernet II, Src: PcsCompu_f8:60:55 (08:00:27:f8:60:55), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Destination: Broadcast (ff:ff:ff:ff:ff:ff) > Source: PcsCompu_f8:60:55 (08:00:27:f8:60:55) > Type: 802.1Q Virtual LAN (0x8100) > 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 14 > Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255 > User Datagram Protocol, Src Port: 68, Dst Port: 67 > Dynamic Host Configuration Protocol (Discover)							

ההודעה עוברת גם מ-S2 ל-3PC:

709	16:46:27.873359	0.0.0.0	255.255.255.255	DHCP	342	0 ✓	DHCP Discover
-----	-----------------	---------	-----------------	------	-----	-----	---------------

Wireshark - Packet 709 - router_vlan_pc3.pcapng							
> Frame 709: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0 > Ethernet II, Src: PcsCompu_b8:e6:de (08:00:27:b8:e6:de), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255 > User Datagram Protocol, Src Port: 68, Dst Port: 67 > Dynamic Host Configuration Protocol (Discover)							

ההודעה עוברת גם מ-S2 ל-4PC:

695	16:46:27.873359	0.0.0.0	255.255.255.255	DHCP	342	0 ✓	DHCP Discover
-----	-----------------	---------	-----------------	------	-----	-----	---------------

Wireshark - Packet 695 - router_vlan_pc4.pcapng							
> Frame 695: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface -, id 0 > Ethernet II, Src: PcsCompu_b8:e6:de (08:00:27:b8:e6:de), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255 > User Datagram Protocol, Src Port: 68, Dst Port: 67 > Dynamic Host Configuration Protocol (Discover)							

6.13 המחשבים באותו לאן ילכו ישירות אחד לשני, כאשר מחשבים בלנים שונים יעברו בדרך אצל הראוטר והוא ידע להפנות אותם ולהעביר את ההודעות בין הלאנים.